

A Radius–Minimum-Degree Bound for Induced Trees and Graffiti.pc Conjecture 84

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Abstract

For a finite connected graph G , let $t(G)$ denote the largest order of an induced subtree, let $r(G)$ denote the radius, and let $\delta(G)$ denote the minimum degree. Graffiti.pc Conjecture 84 asserts

$$t(G) \geq \frac{2r(G)}{\delta(G)}.$$

We prove the equivalent integral inequality

$$2r(G) \leq t(G)\delta(G).$$

The case $\delta(G) \geq 2$ follows immediately from the Erdős–Saks–Sós induced-path bound $t(G) \geq 2r(G) - 1$. The only substantive case is $\delta(G) = 1$. We prove the stronger leaf lemma: every finite connected graph of radius r having a leaf contains an induced tree on at least $2r$ vertices. The proof uses a minimal counterexample, the radius-drop characterization of non-cut vertices, the structure theorem for vertex-radius-decreasing graphs with cut vertices, and a rooted form of Fan Chung’s proof of the $2r - 1$ induced-path theorem.

1 Definitions and classical ingredients

All graphs are finite, simple, and undirected. For a connected graph G , write

$$\text{ecc}_G(v) = \max_{x \in V(G)} d_G(v, x), \quad r(G) = \min_{v \in V(G)} \text{ecc}_G(v).$$

A vertex c is *central* if $\text{ecc}_G(c) = r(G)$. A vertex x is the *unique eccentric point* (UEP) of c if $d_G(c, x) = \text{ecc}_G(c)$ and every $y \neq x$ satisfies $d_G(c, y) < \text{ecc}_G(c)$.

Let $t(G)$ be the maximum cardinality of a vertex set inducing a tree. A vertex is a *non-cut vertex* if its deletion leaves the graph connected.

We need the following classical radius-drop criterion.

Proposition 1.1 (Radius-drop criterion). *Let G be connected and let v be a non-cut vertex. Then*

$$r(G - v) < r(G)$$

if and only if v is the unique eccentric point of some central vertex of G . In this case

$$r(G - v) = r(G) - 1.$$

This criterion goes back to work of Gliviak and Fajtlowicz; see Swart [4, Proposition 2.2.1].

A nontrivial graph G is called *vertex-radius-decreasing* (vrd) if $r(G - v) < r(G)$ for every non-cut vertex v .

For $\ell \geq 1$, let $S_\ell(B)$ be the graph obtained by attaching to every vertex of B a new pendant path of length ℓ , using that vertex as one endpoint of the path.

Theorem 1.2 (Structure of vrd graphs with a cut vertex). *If G is vrd and has a cut vertex, then*

$$G = S_\ell(B)$$

for some $\ell \geq 1$ and some vrd block B . Moreover the center of G induces B , and

$$r(G) = r(B) + \ell.$$

Conversely every such $S_\ell(B)$ is vrd.

This is Theorem 2.2.10 in Swart [4], where it is noted to follow from a theorem of Gliviak [3]; the local ingredient that every non-cut vertex of a vrd graph with a cut vertex has degree one is attributed there to Fajtlowicz [2]. We also use the standard consequence that the only vrd graph of radius one is K_2 .

The second classical ingredient is the induced-path theorem.

Theorem 1.3 (Erdős–Saks–Sós; Chung). *Every connected graph G of radius r contains an induced path on at least $2r - 1$ vertices. Consequently,*

$$t(G) \geq 2r - 1.$$

See Erdős, Saks, and Sós [1, Theorem 2.2]. Their published proof was supplied by Fan Chung.

We shall need the following rooted version. It is not a new argument; it is the same Chung construction with the radius-decreasing vertex prescribed.

Lemma 1.4 (Rooted Chung lemma). *Let F be connected with radius $r \geq 2$, and let a be a non-cut vertex such that*

$$r(F - a) = r - 1.$$

Then F contains an induced path on at least $2r - 1$ vertices that contains a .

Proof. Let v_0 be a center of $F - a$. Since every vertex other than a is at distance at most $r - 1$ from v_0 , whereas $r(F) = r$, we have $d(v_0, a) = r$. Fix a geodesic

$$v_0, v_1, \dots, v_r = a.$$

Since every vertex has eccentricity at least r , choose w with $d(v_2, w) \geq r$. Necessarily $w \neq a$, and hence

$$r - 2 \leq d(v_0, w) \leq r - 1.$$

Let P be a v_0 – w geodesic.

We recall the chord argument from Chung’s proof. If a vertex u of P were adjacent to v_j for some $j \geq 2$, then

$$\begin{aligned} d(v_0, w) &= d(v_0, u) + d(u, w) \\ &\geq d(v_0, v_j) - 1 + d(v_j, w) - 1 \\ &\geq d(v_0, v_j) - 2 + d(v_2, w) - d(v_2, v_j) \\ &\geq r, \end{aligned}$$

because the fixed v_0 - a path is geodesic and hence $d(v_0, v_j) - d(v_2, v_j) = 2$. This contradicts $d(v_0, w) \leq r - 1$. Thus the concatenation

$$v_r, v_{r-1}, \dots, v_1, v_0, P$$

has $2r - 1$ or $2r$ vertices and has no possible chord except one joining v_1 to the first vertex of P after v_0 . Exactly as in the published proof, if that exceptional chord occurs, deleting v_0 leaves an induced path of order at least $2r - 1$. The prescribed vertex $v_r = a$ is never deleted. Hence the resulting induced path contains a . $\square$

2 The leaf lemma

Lemma 2.1 (Leaf lemma). *Let G be a finite connected graph with a leaf. If $r = r(G)$, then*

$$t(G) \geq 2r.$$

Proof. Suppose otherwise, and choose a counterexample G of minimum order. Fix a leaf u , let a be its unique neighbor, and write $r = r(G)$. If $r = 1$, the edge ua itself is an induced tree of order $2 = 2r$, a contradiction. Hence $r \geq 2$. In particular $|V(G)| \geq 4$, and a is a cut vertex.

Claim 1. For every non-cut vertex $v \neq u$,

$$r(G - v) = r - 1. \tag{1}$$

Indeed $v \neq a$, because a is a cut vertex. Therefore $G - v$ remains connected and u remains a leaf. If $r(G - v) \geq r$, minimality of G would give an induced tree in $G - v$ of order at least $2r(G - v) \geq 2r$, which is also induced in G , a contradiction. Thus $r(G - v) < r$, and Proposition 1.1 gives (1).

Set

$$H = G - u.$$

Deleting a leaf does not change distances between the remaining vertices. A leaf cannot be a center of a connected graph of order at least three, so a center of G remains in H and has eccentricity at most r . Consequently $r(H) \leq r$. If $r(H) < r$, Proposition 1.1, applied to the non-cut vertex u , gives $r(H) = r - 1$. Hence

$$r(H) \in \{r - 1, r\}. \tag{2}$$

Case A: $r(H) = r - 1$

By (1), deletion of every non-cut vertex other than u lowers the radius, and deletion of u lowers the radius by the assumption of this case. Thus G is vrd. Since a is a cut vertex, Theorem 1.2 gives

$$G = S_\ell(B), \quad \ell \geq 1,$$

where B is a vrd block. Put $s = r(B)$. Then

$$r = s + \ell. \tag{3}$$

If $s = 1$, then $B = K_2$, so $G = S_\ell(K_2)$ is the path $P_{2\ell+2} = P_{2r}$ and we are done.

Assume $s \geq 2$. By Theorem 1.3, B contains an induced path Q on at least $2s - 1$ vertices. In $S_\ell(B)$, retain every vertex of Q together with every vertex on the ℓ -edge pendant path attached to

a vertex of Q . Since the pendant paths are mutually internally disjoint and have no edges between them, the induced subgraph on these vertices is a tree. Its order is at least

$$(2s - 1)(\ell + 1).$$

Using $s \geq 2$ and $\ell \geq 1$,

$$\begin{aligned} (2s - 1)(\ell + 1) - 2(s + \ell) &= \ell(2s - 3) - 1 \\ &\geq 0. \end{aligned}$$

Together with (3), this gives an induced tree of order at least $2r$, a contradiction.

Case B: $r(H) = r$

First, H has no leaf. Otherwise H would be a smaller connected graph with a leaf and radius r , so minimality would yield $t(H) \geq 2r$, again a contradiction. Therefore

$$\delta(H) \geq 2. \tag{4}$$

Claim 2. For every non-cut vertex $v \neq a$ of H ,

$$r(H - v) = r - 1. \tag{5}$$

Since $v \neq a$, the graph $G - v$ is obtained from the connected graph $H - v$ by attaching the leaf u at a , so v is also a non-cut vertex of G . By Claim 1,

$$r(G - v) = r - 1.$$

The graph $G - v$ has at least three vertices, hence its leaf u is not central. Deleting u from $G - v$ therefore leaves a graph $H - v$ whose radius is at most $r - 1$. Since $r(H) = r$, deletion of v lowers the radius of H , and Proposition 1.1 gives the exact equality (5).

Claim 3. The graph H has no cut vertex.

Suppose that H has a cut vertex and choose an end block B_0 , with c its unique cut vertex. By (4), B_0 is not a bridge block K_2 ; hence c has at least two neighbors in B_0 . Choose a neighbor $v \neq a$ of c in B_0 . Then v is non-cut in H and $\deg_H(v) \geq 2$. By Claim 2 and Proposition 1.1, v is the unique eccentric point of a central vertex z of H .

We use a local observation. If v is the UEP of a central vertex z and $\deg(v) \geq 2$, then every neighbor of v is non-cut. Indeed, let x be a neighbor of v and choose another neighbor $y \neq x$ of v . Since $d(z, v) = r$ and v is the unique vertex at distance r from z , we have $d(z, x) = r - 1$. For any $q \notin \{v, x\}$, no z - q geodesic can pass through x : such a geodesic would have length at least r , while $q \neq v$ implies $d(z, q) \leq r - 1$. In particular a z - y geodesic avoids x , and appending the edge yv connects v to z in $H - x$. Every other vertex also connects to z in $H - x$. Thus $H - x$ is connected.

Taking $x = c$ contradicts that c is a cut vertex. Hence Claim 3 holds.

By Claim 3 every vertex of H is non-cut. For each $v \neq a$, Claim 2 and Proposition 1.1 give a central vertex c_v whose unique eccentric point is v . Distinct v give distinct c_v , since a fixed vertex cannot have two different unique eccentric points. Thus, if $n = |V(H)|$, at least $n - 1$ vertices of H are central.

Claim 4. The graph H is self-centered.

There is at most one noncentral vertex. If such a vertex b existed, every $y \neq b$ would be central, so

$$d_H(y, b) \leq r.$$

By symmetry $d_H(b, y) \leq r$ for every $y \neq b$, and therefore $\text{ecc}_H(b) \leq r$. Since $r(H) = r$ always implies $\text{ecc}_H(b) \geq r$, we obtain $\text{ecc}_H(b) = r$, contradicting that b is noncentral. Hence every vertex is central.

Now form an auxiliary simple graph D on $V(H)$ by

$$xy \in E(D) \iff d_H(x, y) = r.$$

Since H is self-centered and $r > 0$, every vertex has a vertex at distance r , so

$$\delta(D) \geq 1. \tag{6}$$

For each $v \neq a$, the vertex c_v is adjacent to v in D and has degree exactly one in D . Therefore every vertex $v \neq a$ has a degree-one neighbor in D .

We need the following elementary finite observation.

Lemma 2.2. *Let D be a finite simple graph with minimum degree at least one, and let $a \in V(D)$. If every vertex $v \neq a$ has a neighbor of degree one, then a also has a neighbor of degree one.*

Proof. Suppose not. Choose $x \in N_D(a)$ using $\delta(D) \geq 1$. Since a has no degree-one neighbor, $\deg_D(x) \geq 2$. Because $x \neq a$, choose a degree-one neighbor y of x .

If $y \neq a$, apply the hypothesis to y . Its only neighbor is x , so x must have degree one, a contradiction. Hence $y = a$, and $\deg_D(a) = 1$. Since $\deg_D(x) \geq 2$, choose $z \in N_D(x) \setminus \{a\}$. The vertex $z \neq a$ has a degree-one neighbor w . Since a is adjacent only to x , we have $w \neq a$; also $w \neq x$, because $\deg_D(x) \geq 2$. Applying the hypothesis to the degree-one vertex w forces its only neighbor z to have degree one. But z is adjacent to the two distinct vertices x and w , a contradiction. $\square$

By Lemma 2.2, choose c adjacent to a in D with $\deg_D(c) = 1$. Thus

$$d_H(c, a) = r$$

and a is the unique eccentric point of c . By Claim 4, c is central. Proposition 1.1 therefore gives

$$r(H - a) = r - 1. \tag{7}$$

Claim 3 says a is non-cut. Apply the rooted Chung lemma to H and a . We obtain an induced path P on at least $2r - 1$ vertices containing a . Finally, reinsert the original leaf u . Since u is adjacent in G only to a , the induced subgraph

$$G[V(P) \cup \{u\}]$$

is a tree on at least $2r$ vertices. This is the final contradiction. $\square$

3 Graffiti.pc Conjecture 84

Theorem 3.1. *Let G be a finite connected graph with at least two vertices. Then*

$$2r(G) \leq t(G)\delta(G).$$

Equivalently,

$$t(G) \geq \frac{2r(G)}{\delta(G)}.$$

Proof. A connected nontrivial graph has $\delta(G) \geq 1$ and $r(G) \geq 1$.

If $\delta(G) = 1$, then G has a leaf. Lemma 2.1 gives $t(G) \geq 2r(G)$, hence

$$t(G)\delta(G) = t(G) \geq 2r(G).$$

Now suppose $\delta(G) \geq 2$. By Theorem 1.3,

$$t(G) \geq 2r(G) - 1.$$

Therefore

$$\begin{aligned} t(G)\delta(G) &\geq 2(2r(G) - 1) \\ &= 4r(G) - 2 \\ &\geq 2r(G), \end{aligned}$$

where the last inequality uses $r(G) \geq 1$. □

Remark 3.2. The proof deliberately avoids the tempting but false strengthening that a graph with a leaf must contain an induced path on $2r(G)$ vertices. The leaf lemma is genuinely a statement about induced trees: the extra vertex in Case B is attached as a pendant vertex to a long induced path, and in Case A the required tree is obtained from several pendant branches over an induced path in the vrd core.

4 Formalization notes

For Lean 4 formalization, the proof naturally separates into the following modules:

- (i) the Erdős–Saks–Sós/Chung induced-path kernel, including the rooted variant Lemma 1.4;
- (ii) the radius-drop criterion, UEP definitions, and elementary deletion lemmas;
- (iii) the vrd cut-vertex structure theorem, $G = S_\ell(B)$;
- (iv) the minimal-counterexample proof of Lemma 2.1;
- (v) the final arithmetic reduction in Theorem 3.1.

The separate Lean package accompanying this manuscript keeps (iii) as an explicit formalization target rather than hiding it behind an axiom.

References

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